HW3 – Creating your custom ‘Car’ class and implementing iterators allowing to control the speed of the ‘Car’

1) Create “**HW3.ipynb**” – this will be the file for your homework.
2) In this home task you will create your own **Car** class, which has its own state (on/off the road) and is able to:
    a. Increase speed to some value;
    b. Decrease speed to some value;
    c. Go to parking (get off the road);
    d. Show the current weather;
    e. Show, how many cars are on the road right now.
3) You will practice here OOP (classes, @staticmethod, @classmethod, association) and FP (writing custom iterators to understand how they work).
4) The description of the methods is provided within **car-snippet.py**
5) The weather API code is provided in **weather-snippet.py**
6) Your task is to:
    a. Complete two iterator classes that will control the speed (increase and decrease). We will put them into our Car class by association approach. On the last lecture we have covered Cat and Toy example, which was <u>Aggregation type</u> of association (remember, Toys were created separately from our Cats, and in principle could be utilized by any other Cat as well). Now, we will use <u>Composition type</u> of association. Our iterator classes will be initialized inside the Car class to be bound to the specific Car instance. Thus, our iterators will have access to the state of the car and its current speed.
    b. Complete Car class. The methods you have to define within this class are described in **car-snippet.py.** There are two methods, where you have a detailed description, and three methods for which you have to decide yourself: which one is regular, which is **@staticmethod** and which is **@classmethod** (and to add corresponding decorators where needed).
7) Create “**HW3**” branch in your repo.
8) Create a directory “**HW3**” inside.
9) Upload your IPYNB file into “HW3” and create a pull request to submit your work.

Soft Deadline for this task is 19.04.2026 23.59 MSK. After this date, **half** points will be graded for the task.

Hard Deadline for this task is 26.04.2026 23.59 MSK. After this date, **no** points will be graded for the task.